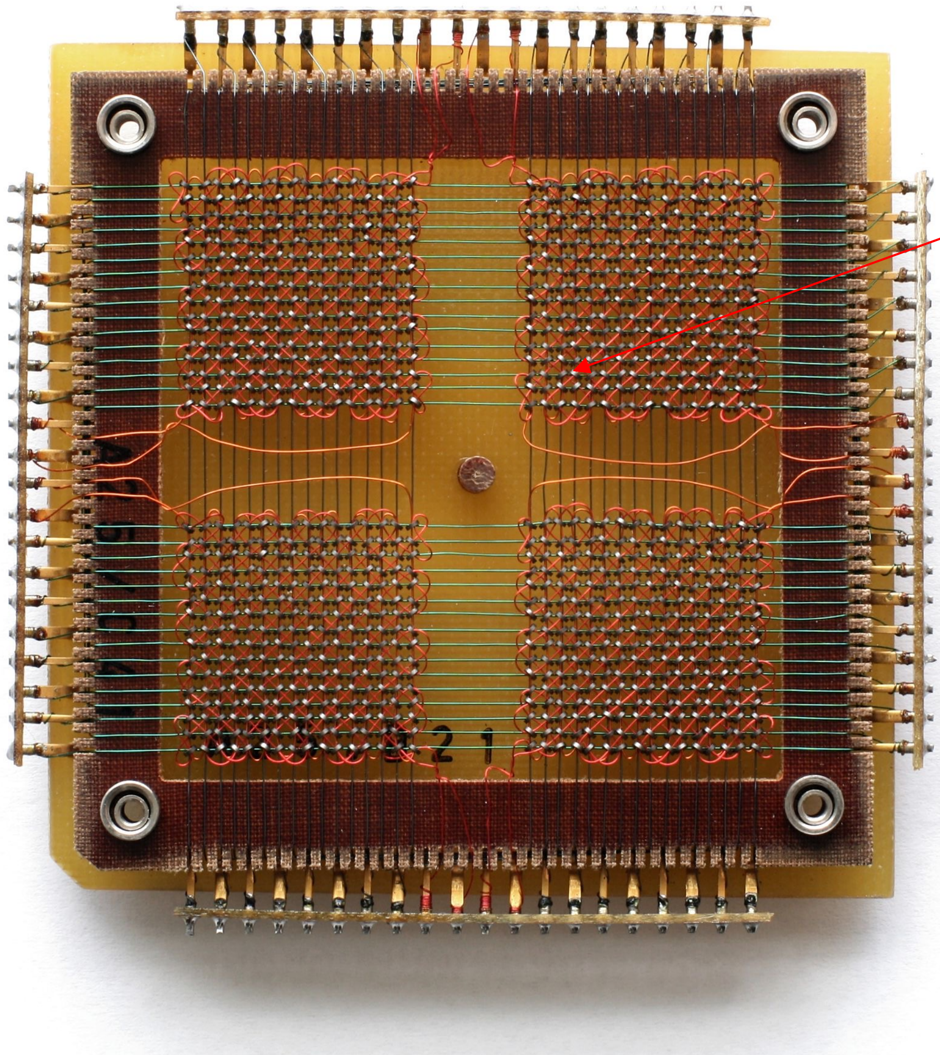
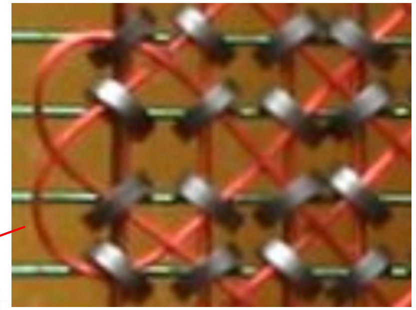
Tutorial 7

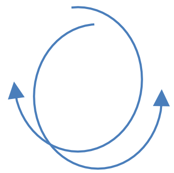
SC1006, CZ1106

Semiconductor Memories, HDD and SSD

Magnetic-core memory was the predominant form of random-access computer memory for 20 years between about 1955 and 1975. Such memory is often just called core memory, or, informally, core.







Each ring could be magnetized in two directions, hence could store 0 or 1 **even when the power was removed**

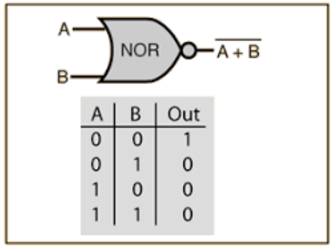
A 32 x 32 core memory plane storing 1024 bits (or 128 bytes) of data. The small black rings at the intersections of the grid wires, organised in four squares, are the ferrite cores.

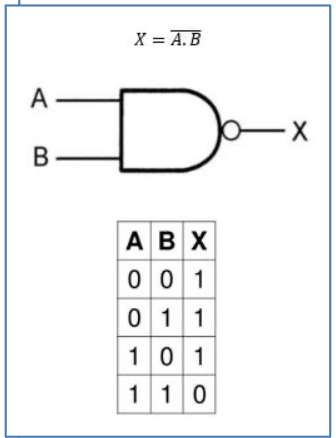
System and Storage Memory

DRAM - Dynamic random-access memory is a type of random-access semiconductor memory that stores each bit of data in a memory cell, usually consisting of a tiny capacitor and a transistor, both typically based on metal-oxide-semiconductor (MOS) technology.

SRAM - Static random-access memory is a type of random-access memory that uses latching circuitry (flip-flop) to store each bit. SRAM is volatile memory; data is lost when power is removed.

Flash memory is an electronic non-volatile computer memory storage medium that can be electrically erased and reprogrammed. The two main types of flash memory, **NOR flash** and **NAND flash**, are named for the NOR and NAND logic gates.





A hard disk drive (HDD), hard disk, hard drive, or fixed disk is an electro-mechanical data storage device that stores and retrieves digital data using magnetic storage and one or more rigid rapidly rotating platters coated with magnetic material.

A solid-state drive (SSD) is a solid-state storage device that uses integrated circuit assemblies to store data persistently, typically using flash memory, and functioning as secondary storage in the hierarchy of computer storage. It is also sometimes called a semiconductor storage device, a solid-state device or a solid-state disk, even though SSDs lack the physical spinning disks and movable read–write heads used in hard disk drives (HDDs) and floppy disks.

System and Storage Memory

1. What is the difference between system and storage memory? Elaborate your answer with an example of a person working on his Laptop Computer, using a Power Point application from the Microsoft Office suite to draft his presentation slides.

System memory is the memory where the CPU executes the code and accesses the working copy of data during runtime.

Storage memory is where the computer stores the entire set of program and data. Majority of the time, storage memory does not support code execution.

In the example of a laptop, the system memory typically corresponds to the DRAM connected to the processor. The entire Microsoft Office Suite will be too large to be stored in the system memory, so it is likely to be stored in the Storage memory, which is the HDD or SSD in the laptop. During runtime, part of the power point application is loaded into the system for execution. Any data (photos, graphics, video, etc.) that is required to assemble the .ppt slide would also need to be loaded into the system memory before the processor could perform any operation on it. Note that the data needs not be the entire photo/video, only the subset that the processor works on need-to-be in the system memory at an instance. The final .ppt file is stored in the storage memory.

Semiconductor Memories

2. What memory type (SRAM, DRAM, NOR Flash, NAND Flash, etc.) would you use to implement the cache in Processor Cx1006-200M16 in the case study notes? Explain your choice.

SRAM. Cache needs to be implemented with very fast memory so as to keep up with the processor operating speed. DRAM is too slow and its transfer procedure is too complex to be used for cache operation. Flash is not suitable due to its finite erase cycles and slower speed.

SRAM - Static random-access memory is a type of random-access memory that uses latching circuitry (flip-flop) to store each bit. SRAM is volatile memory; data is lost when power is removed.
DRAM - Dynamic random-access memory is a type of random-access semiconductor memory that stores each bit of data in a memory cell, usually consisting of a tiny capacitor and a transistor, both typically based on metal-oxide-semiconductor (MOS) technology.
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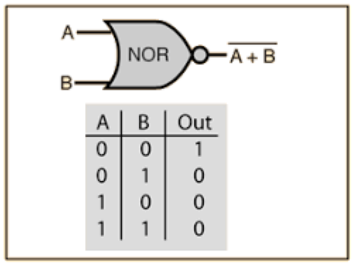

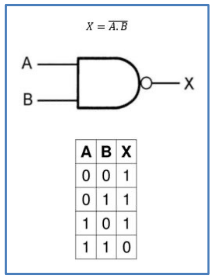

Semiconductor Memories

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SRAM vs. DRAM

STATIC RAM	DYNAMIC RAM
Does not require refreshing	Must be continuously refreshed
Requires multiple transistors to store one bit	Requires one transistor and one capacitor to store one bit
Delivers faster access times	Delivers slower performance
Consumes less power, especially in idle	Consumes more power
Takes up more space	Requires less space
Can hold only a small amount of data	Can hold much more data
Costs more than DRAM	Costs less than SRAM
Typically used for a processor's cache	Typically used for a computer's main memory

Semiconductor Memories

3. Name two main types of Flash memory available in the market. What are the differences between them? Which application/product areas are they used in?

NAND and NOR Flash Memory.

NAND:

- Data is accessed a page at a time. There are commands that are used to open a particular page followed by the individual bytes read/write. Does not support Execute in Place.
- More compact due to layout of the cells => more cost effective.
- Used mainly as storage memory. E.g., SSD, USB Flash Drive.

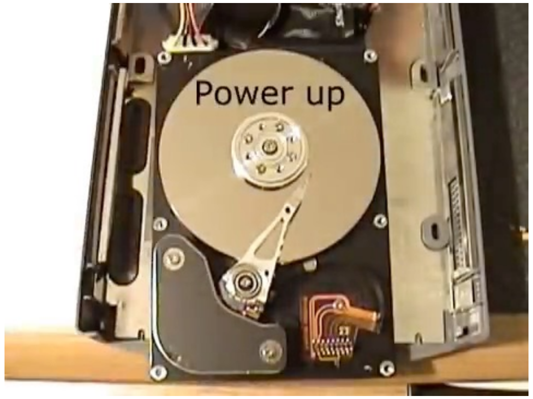
NOR:

- Allows Random Data Read by providing address information only.
- Supports Execute in Place, i.e. Code stored in NOR Flash can be executed directly without having to transfer to RAM first.
- Less compact than NAND flash.
- Used mainly as system memory to store program code.

HDD

4. Reference the two HDDs listed in the case study notes (HDD001 and HDD002)

   a. What is the **capacity** of each drive?



(h) Part Number: HDD001
- 4 surfaces
- 1024 tracks per surface
- 128 sectors per track
- 512 bytes/sector
- Track-to-track seek time = 5 msec
- Rotational speed = 5000 RPM
- MTTF = 500,000 hours

(i) Part Number: HDD002
- 8 heads
- 1024 cylinders
- 256 sectors per track
- 512 bytes/sector

A hard disk drive (HDD), hard disk, hard drive, or fixed disk is an electro-mechanical data storage device that stores and retrieves digital data using magnetic storage and one or more rigid rapidly rotating platters coated with magnetic material.

Capacity: Decimal vs. Binary

Everyone who has ever purchased a hard drive finds out the hard way that there are two ways to define a gigabyte.

When you buy a "500 Gigabyte" hard drive, the vendor defines it using the decimal powers of ten definition of the "Giga" prefix.

$500 * 10^9$ bytes = 500,000,000,000 = 500 Gigabytes

But the operating system determines the size of the drive using the computer's binary powers of two definition of the "Giga" prefix:

$465 * 2^{30}$ bytes = 499,289,948,160 = 465 Gigabytes

giga	GB	109	230	7%	Greek root for giant
tera	TB	1012	240	10%	Greek root for monster
peta	PB	1015	250	13%	Greek root for five, "penta"

If you're wondering where 35 Gigabytes of your 500 Gigabyte drive just disappeared to, you're not alone. It's an old trick perpetuated by hard drive makers-- they intentionally use the official SI definitions of the Giga prefix so they can inflate the the sizes of their hard drives, at least on paper.

HDD

4. Reference the two HDDs listed in the case study notes (HDD001 and HDD002)

   a. What is the **capacity** of each drive?

HDD001:

Capacity = 4*1024*128*512 = 268, 435, 456 Bytes = 256Mbytes.

Note: 1Mbyte = 2^20 bytes

HDD002:

Capacity = 8*1024*256*512 = 1, 073, 741, 824 bytes = 1GByte.

Note:

1GByte = 2^30 bytes,

# of cylinders = # of tracks of each surface,

# of heads = # of valid surfaces

(h) Part Number: HDD001

- 4 surfaces
- 1024 tracks per surface
- 128 sectors per track
- 512 bytes/sector
- Track-to-track seek time = 5 msec
- Rotational speed = 5000 RPM
- MTTF = 500,000 hours

(i) Part Number: HDD002

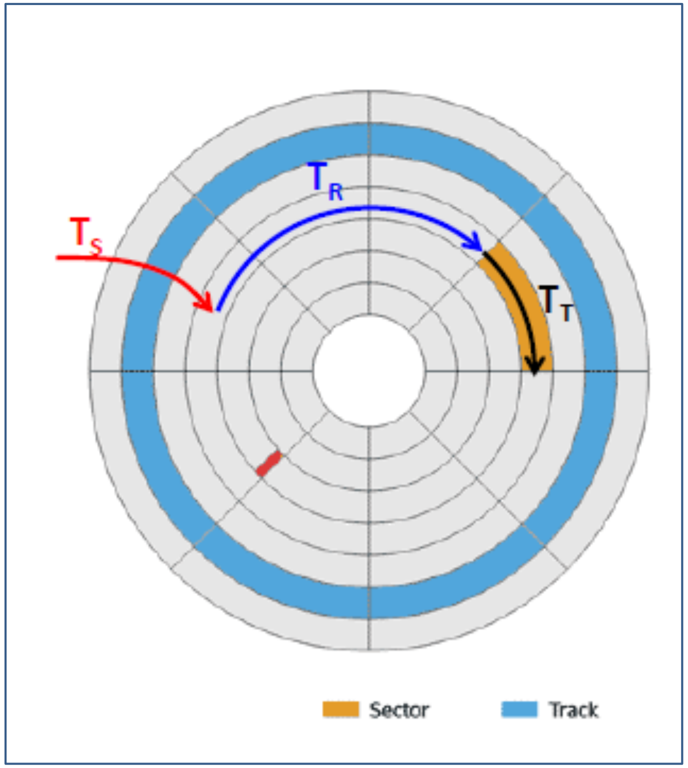
- 8 heads
- 1024 cylinders
- 256 sectors per track
- 512 bytes/sector

HDD

4. Reference the two HDDs listed in the case study notes (HDD001 and HDD002)

   i. What is its **access time**?

(h) Part Number: HDD001
- 4 surfaces
- 1024 tracks per surface
- 128 sectors per track
- 512 bytes/sector
- Track-to-track seek time = 5 msec
- Rotational speed = 5000 RPM
- MTTF = 500,000 hours



RPS = 5000/60

$$T_{R,AV} = \frac{0.5}{RPS} \text{ seconds}$$

Average rotational delay = $T_{R,AV}$ = 0.5/RPS = 0.5/(5000/60) = 6ms

Seek time = T_S = 5ms

Access time, $T_A = T_S + T_{R,AV} = 5 + 6 = 11$ms

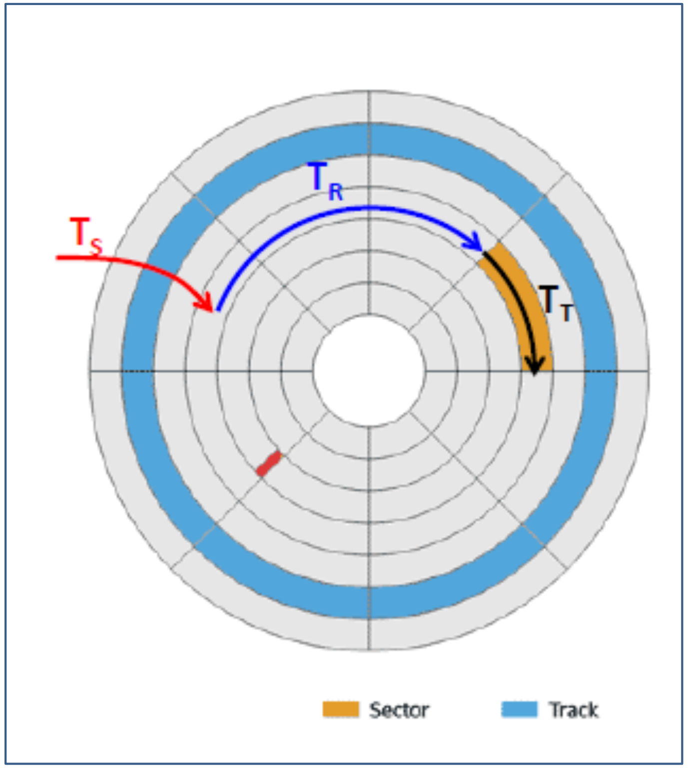
A rotational delay is the time between information requests and how long it takes the hard drive to move to the correct sector.
Seek time is the time taken for a hard disk controller to locate a specific piece of stored data.

HDD

ii. What is the time needed to transfer a 4Kbyte file stored in random non-consecutive sectors on different tracks? Assume that every sector is on a different track.

(h) Part Number: HDD001

- 4 surfaces
- 1024 tracks per surface
- 128 sectors per track
- 512 bytes/sector
- Track-to-track seek time = 5 msec
- Rotational speed = 5000 RPM
- MTTF = 500,000 hours



$$T_{R,AV} = \frac{0.5}{RPS} \text{ seconds}$$

$$T_T = \frac{N}{RPS * D_T * D_S}$$

RPS = 5000/60 per second

Access time T_A = 11 ms

\# of sectors = 4*1024/512 = 8

Transfer time T_T for 1 sector = 512/(RPS*D_T*D_S) = 512/((5000/60)*128*512) = 93.75 us

Total time, T_{TOTAL} = 8*(T_A + T_T) = 88.75 ms

HDD

iii. After defragmenting HDD001, what would be the time needed to transfer a 280Kbyte file?

Defragmentation => file is stored in consecutive sectors.

However, 280KByte is larger than the size of one track (128*512=64KByte).

It occupies 4 full tracks and 24 Kbyte on the fifth track. 4*64+24=256+24

RPS = 5000/60 per second
Access time T_A = 11 ms

T_T for one full track = 1/RPS = 12 ms

Total time for 1 track, T_{TOTAL} = 11+12 ms = 23 ms

4 tracks = 4*23 = 92ms

Last 24KByte:
Transfer time T_T = N/(RPS*DT*DS) = (24*1024)/((5000/60)*128*512) = 4.5 ms
Total time, T_{TOTAL} = 11+4.5ms = 15.5 ms

Total time needed to transfer 280KByte file = 92+15.5 = 107.5 ms

Note: T_A is required when moving from one track to the other since the R/W head need to be positioned at the starting sector of the next track.

HDD and SSD

c. If you are building a Network Access Storage for your home to act as a backup storage for your home's computers, which HDD would you choose? Justify your choice. MTTF in the HDD parameters refer to Mean-Time-To-Failure. It's a statistical approximation of how long a product could last before failing. Note that MTTF=1M hours doesn't mean the product's mean time to failure is 1M hours, but the larger MTTF value does indicate that the product is more reliable (statistically).

(h) Part Number: HDD001

- 4 surfaces
- 1024 tracks per surface
- 128 sectors per track
- 512 bytes/sector
- Track-to-track seek time = 5 msec
- Rotational speed = 5000 RPM
- MTTF = 500,000 hours

(i) Part Number: HDD002

- 8 heads
- 1024 cylinders
- 256 sectors per track
- 512 bytes/sector
- Track-to-track seek time = 4 msec
- Rotational speed =10000 RPM
- MTTF = 1,000,000 hours

HDD002.

- Statistically more robust as MTTF = 1M hours, which is 2X that of HDD001. So less likely to fail.
- Higher storage
- Higher Performance (higher RPM, shorter Seek Time)
- HDD drives for NAS is usually more expensive (per bit wise) than typical consumer drives.

HDD and SSD

If you are building a Network Access Storage for your home

d. Would you use a SSD instead for Q4(c) above? Since SSD is more robust than HDD and robustness is very important for backup storage.

HDD.

- SSD is still has a higher cost per bit than HDD. Not ideal for application that requires huge storage, especially for consumer home use.
- Finite read/write cycles of SSD compared to HDD's almost infinite read/write cycles.
- SSD cannot replace HDD in backup storage area, including data centers and servers, yet. This primarily due to cost.

A solid-state drive (SSD) is a solid-state storage device that uses integrated circuit assemblies to store data persistently, typically using flash memory, and functioning as secondary storage in the hierarchy of computer storage. It is also sometimes called a semiconductor storage device, a solid-state device or a solid-state disk, even though SSDs lack the physical spinning disks and movable read–write heads used in hard disk drives (HDDs) and floppy disks.

Memory types selection

5. What would be the memory choices for the system and storage memory for each scenario below? Justify your memory choice selection in terms of functionality, performance and cost.
   a. Entry level Microsoft Windows desktop computer for general office use but needs huge data storage capacity to store videos relating to the company product.

a. System Memory: DRAM, Storage Memory: Magnetic HDD
   - Entry level Desktop, General office use => low cost and only need average performance => DRAM and HDD as cost is lower than SSD
   - Desktop => No much physical movement expected so robustness to movement is not required.
   - Windows OS typically requires a few GByte of system memory. At this size, SRAM will be too expensive. Higher end product uses higher speed DRAM such as DDR3, DDR4 etc
   - Large storage for videos => Magnetic HDD (lower cost per bit than SSD)

Data Centers

6. The main active storage of Data Centers are HDD. But HDD is prone to crashing due to the mechanical nature of its design. How does these centers mitigate this issue?

- Redundancy.
  - User data is stored in multiple servers at multiple sites. If one breaks down, the other would take over.
  - Within a server, some other error correction/recovery techniques may be used to mitigate localized error.
  - Data can also be backed up to tapes. This serves as the last line of defense.

> To begin with, tape storage is more energy efficient: Once all the data has been recorded, a tape cartridge simply sits quietly in a slot in a robotic library and doesn't consume any power at all. Tape is also exceedingly reliable, with error rates that are four to five orders of magnitude lower than those of hard drives. And tape is very secure, with built-in, on-the-fly encryption and additional security provided by the nature of the medium itself. After all, if a cartridge isn't mounted in a drive, the data cannot be accessed or modified. This "air gap" is particularly attractive in light of the growing rate of data theft through cyberattacks.

7. Why does the Processor Cx1006-200M16 has two different types of non-volatile memory (Flash and EEPROM) on-chip?

Flash has a larger page size (order of Kbytes) so can only be erased at Kbytes level. This is not suitable for certain use case such as storing of user configuration parameters, which is typically done at order of bytes level. An EEPROM, which has a smaller page size (tens of bytes) is more suitable for such use case as it result in less page erases. Also, EEPROM has a larger erase cycle endurance i.e. it could tolerate larger number of erasures before failing.

EEPROM (also E^2PROM) stands for electrically erasable programmable read-only memory and is a type of non-volatile memory used in computers, integrated in microcontrollers for smart cards and remote keyless systems, and other electronic devices to store relatively small amounts of data by allowing individual bytes to be erased and reprogrammed.